DecoPro – Decompaction Workflow

Step-by-Step User Guide

1. Introduction

DecoPro is a software tool designed to calculate the decompaction of sedimentary successions and reconstruct changes in sediment thickness and porosity through time.

The software uses a porosity–depth relationship to account for the progressive reduction in sediment porosity and thickness associated with burial and compaction.

This tutorial provides a step-by-step example of how to prepare the input data, run DecoPro, and interpret the resulting decompaction calculations for an onshore well. The example is based on the dataset presented by Allen and Allen (2013)-Appendix 56, allowing users to compare the software output with published reference values.

2. Reference Dataset

The file **Decompaction_values_allen_allen.csv** contains the original values reported in Allen and Allen (2013).

This dataset is provided as a reference example to demonstrate the decompaction procedure and evaluate the consistency of the results generated by DecoPro.

3. Preparing the Input File

DecoPro requires an input file in CSV (Comma-Separated Values) format.

To prepare the input dataset:

1. Open Decompaction_values_allen_allen.csv.
2. Create a new spreadsheet using the reference dataset.
3. Organize the sedimentary layers according to the required input structure.
4. Ensure that all depth and thickness values are expressed in **kilometres (km)**.

5. Arrange the sedimentary layers from oldest to youngest. The youngest layer should appear in the final row of the dataset.
6. Save the spreadsheet in CSV format. You can choose any file name. In this guide, we named the file “onshore_well_input.csv”. The CSV file must contain the following columns: “id”, “h”, “z2”, “z1”, “c”, and “phi0”.

4. Running DecoPro

Once the input file has been prepared, the decompaction calculation can be performed.

Follow the steps below:

1. **Launch DecoPro:** The DecoPro interface will appear, displaying an input table in the upper section and an Output panel in the lower section.
2. **Load the input file:** Click the **Load From CSV** button and select the previously prepared onshore_well_input.csv file.
3. **Run the calculation:** Once the input data have been verified, click **Submit** to initiate the calculation.
4. **View the results:** The calculated values will appear in the **Output** panel, located below the input table.

5. Understanding the Results

Once the calculation is completed, DecoPro displays the results in the Output panel, located in the lower section of the software interface, directly below the input table.

The results are organized into successive decompaction stages, showing how the sedimentary layers change in thickness, depth, and porosity throughout the decompaction process.

Each stage represents a different step in the reconstruction of the sedimentary succession, beginning with the oldest layer and progressively incorporating the younger overlying layers.

For each stage, DecoPro provides the following information:

Stage: Indicates the successive steps of the decompaction calculation.

Layer base (km): Represents the calculated depth of the base of each sedimentary layer at the corresponding stage.

Average porosity: Represents the calculated average porosity of each layer at that stage, accounting for its burial depth and lithology-dependent compaction parameters.

For example, in Stage 1, the oldest layer (Chalk) has a calculated base depth of 1.6126 km and an average porosity of 0.41682.

In Stage 2, the Sandstone layer is incorporated into the sedimentary succession. The calculated base depth of the Chalk layer increases to 2.444 km, while its average porosity decreases to 0.19197, reflecting the effects of additional burial and compaction.

As successive layers are added, DecoPro calculates the corresponding changes in layer thickness, burial depth, and porosity.

These results allow users to evaluate the progressive compaction of individual sedimentary layers and reconstruct their thickness and porosity evolution through time.

All calculated depths are expressed in kilometres (km), whereas porosity values are expressed as fractions between 0 and 1.

6. Applying DecoPro to New Datasets

After completing the reference example, users can apply the same workflow to their own geological datasets.